User Interview Summary – Precision Seeder Project

1. User Profile

Name: Frank

Occupation: Farmer and professor

Farm size: ~4 hectares

Field location: Minimum 1 km from village (tool must be carried)

2. Current Planting Process (As-Is System)

Ridge preparation: Done October–November.

Digging holes: Using hoe (~$7). Hole depth 10–30 cm. Soil can be heterogeneous and very clayey/sticky.

Seed placement: Done manually by hand. Seeds carried in bags (1–10 kg) or pockets.

Covering: Done using foot. Must avoid drying and avoid compressing ridge.

3. Agronomic Requirements

Planting depth: 10–30 cm (must not exceed rooting zone).

Spacing: 1 seed every 2.5 m.

Fertilizer: 5–8 g per planting station. Must not contact seed (burn risk). Fertilizer hole must be covered.

Replanting required 2–3 days later if germination fails (must avoid this).

Field should ideally be planted in one day to avoid soil moisture variation.

4. Quantitative Data

Seeds required: 20 kg per 2 Ha.

Seed bag weight: 1–10 kg.

Fertilizer per hole: 5–8 g.

Hoe blade width: 7–10 cm.

Hoe handle length: 1–1.5 m.

Willing to pay: $5–7.

Current equipment cost: ~$7.15 total.

5. Physical & Ergonomic Issues

Back pain due to leaning forward.

Arm fatigue from carrying seeds.

High force required in clay soils.

Walking between ridges to avoid damage.

Tool must be carried 1 km to field.

Weight and portability are major concerns.

6. Environmental & Soil Conditions

Low organic matter.

Heterogeneous soil.

Clay can be extremely sticky.

Water is scarce → cleaning must require minimal water.

7. Design Requirements

Accurate and adjustable hole depth (10–30 cm).

Precise seed placement.

Proper covering without compressing ridge.

Fertilizer dosing: 5–8 g, separated from seed.

Adjustable for different crops.

Operable by 1 person (2 acceptable).

Fully manual.

Mechanically simple and durable.

Easy to repair (standard parts, weldable).

Lightweight, compact, possibly foldable.

Easy to clean.

8. Economic Constraints

Willing to spend: $5–7.

Current system: Hoe + bag (~$7.15).

New tool must justify cost replacement.

9. User Priorities

Precision more important than speed.

Reliability is critical.

Avoid failures that require replanting.

10. Main Pain Points

Digging correct depth.

Covering properly.

Avoiding fertilizer-seed contact.

Avoiding ridge compression.

Working in sticky clay.

11. Design Opportunities

Depth-controlled spike mechanism.

Lever-assisted extraction in clay.

Separate fertilizer channel.

Controlled non-compacting covering mechanism.

Integrated seed container.

Lightweight tubular frame construction.

Foldable design for storage and transport.

12. Critical Constraints Summary

Under $7 cost target.

Manual operation.

Lightweight and portable.

Durable and repairable locally.

Works in sticky clay.

Depth-controlled and precise.

Non-compacting covering.

Fertilizer-separated.

Cleanable with minimal water.